Milestone 1

☑ Checkbox	☐

Breathing Intelligence: Morphological Computation in Inflatable Robot Systems

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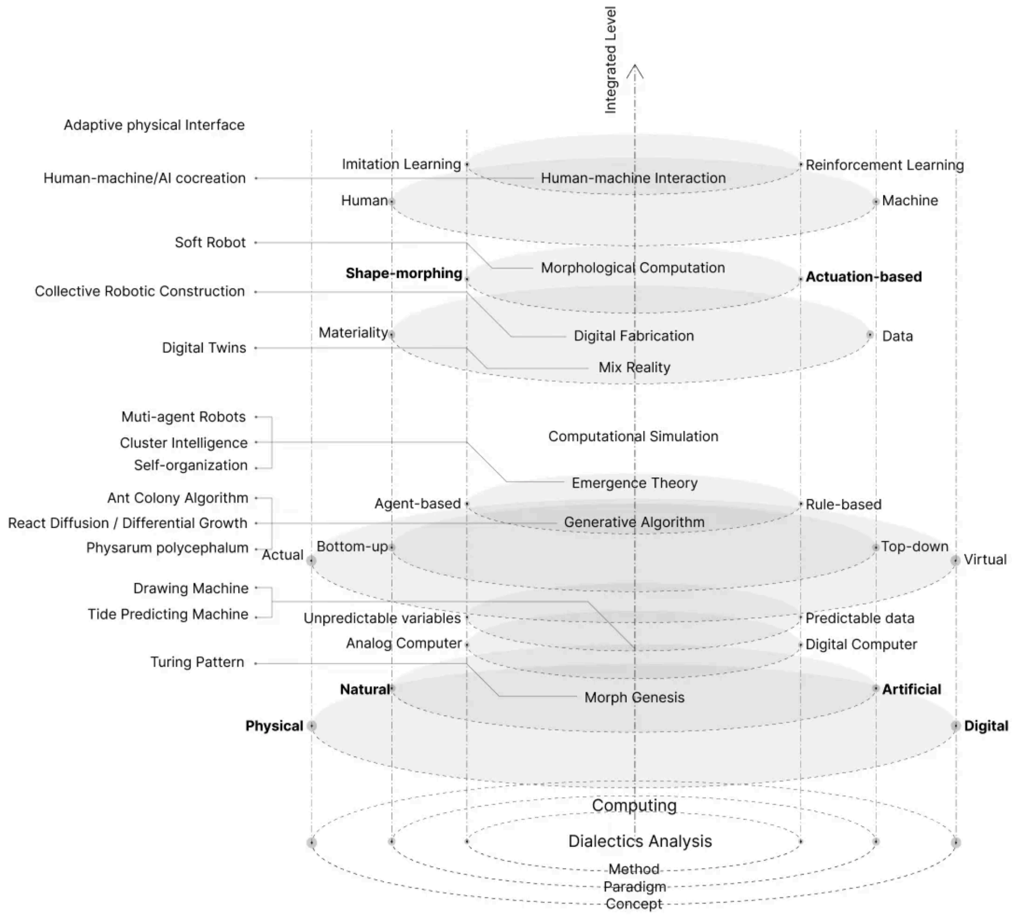
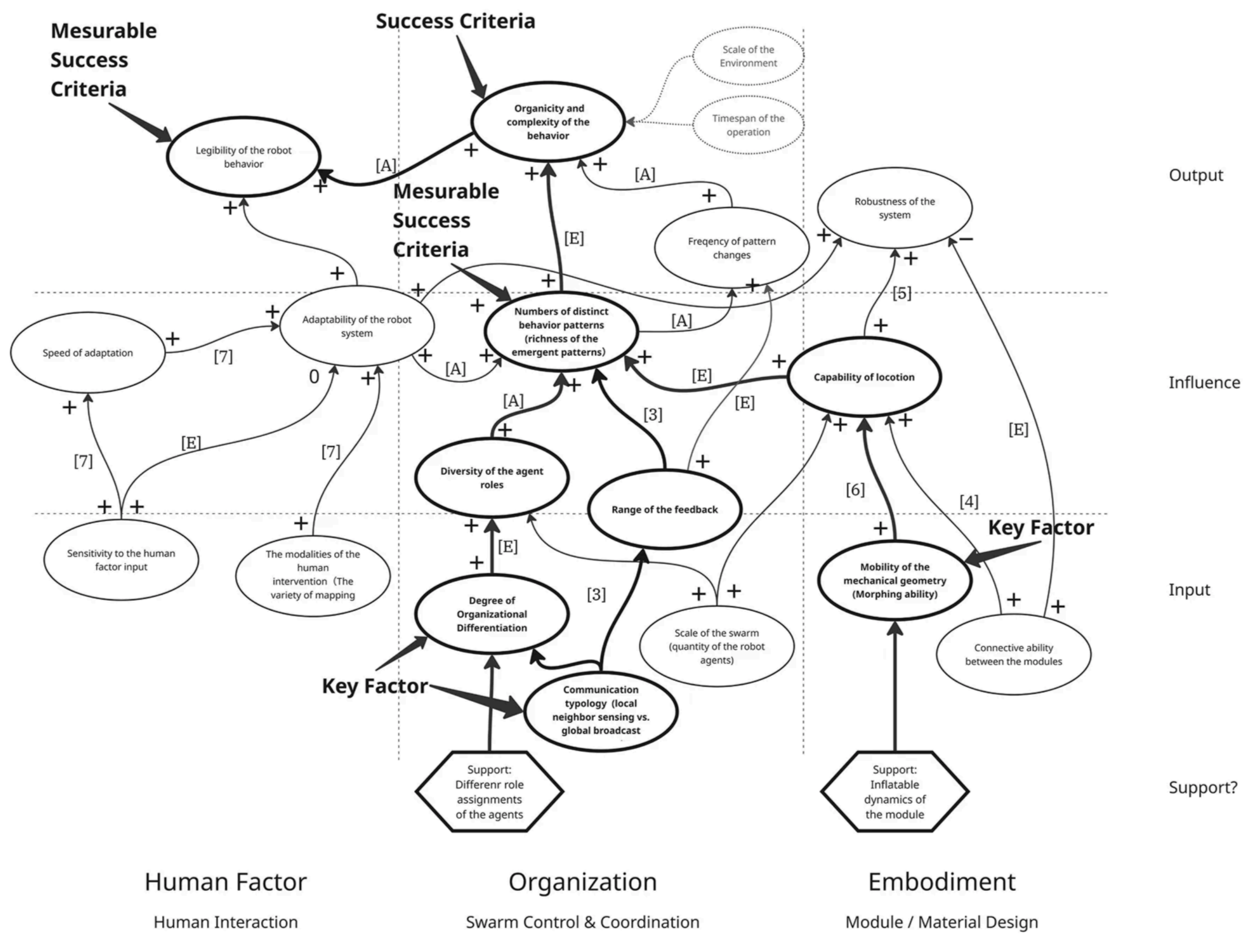
Context and background

Dialectics conceptual mapping (principles)



Impact Model

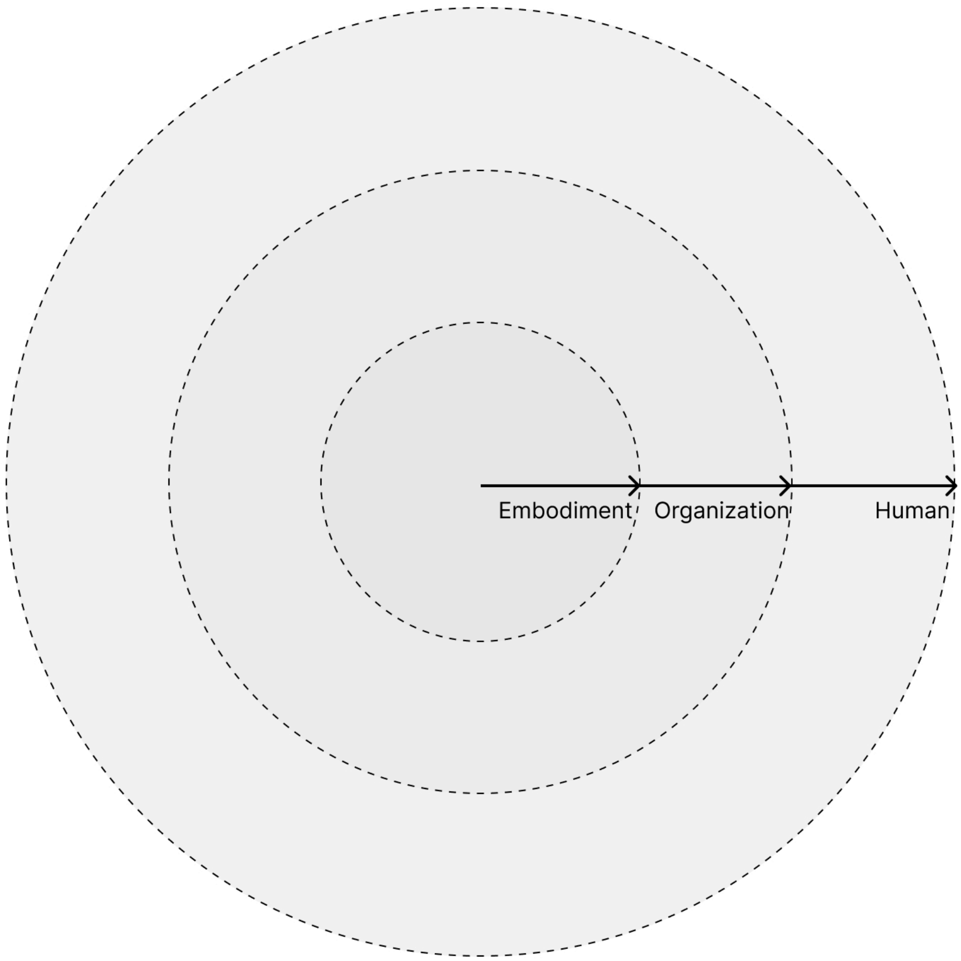
Robot —— Behavior



[1] S. Tibbits. *Things Fall Together: A Guide to the New Materials Revolution*. Princeton University Press, 2021.
[2] Chin, T. *Function Follows Form: An Exploration of Robotic Embodiment through Geometry*, 2023.
[3] *Recent Trends of Multi-Agent AI Systems in Robotics Control — by Swarms*, Medium article .
[4] Camazine, Scott, et al. *Self-Organization in Biological Systems*. Princeton University Press. 2009
[5] Shaoul, S., et al. "Multi-Robot Motion Planning with Diffusion Models." 2025.
[6] Zachary Yoder et al. ,Hexagonal electrohydraulic modules for rapidly reconfigurable high-speed robots.Sci. Robot.9,eadl3546(2024).
[7] Eguchi, R., et al. 2025. FluidicSwarm: Embodiment of Swarm Robots Using Fluid Behavior Imitation. In ACM SIGGRAPH 2025 Emerging Technologies (SIGGRAPH '25). Association for Computing Machinery, New York, NY, USA, Article 5, 1–2.
[8] "Stephen Wolfram: A New Kind of Science | Online—Table of Contents." Accessed November 12, 2025.

Factor clusters — From the center to the peripheral

Exploration steps



Research Questions:

High Level:

Take the soft robot as the “site”, research on the process from simplicity to complexity, from the single to the plural...

Implemental:

How to design a modular robot system that can locomote by deforming?

How to coordinate/control the robot system to enrich the behavior patterns?

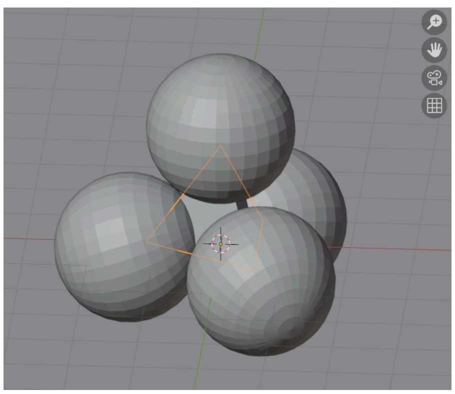
Not problem-driven but more experiment oriented?

→The experiment that I have been working on.

Simulation as an initial validation

Low fidelity simulation in Blender:

1. **Overall robot module design:**



2. **Shape-changing mechanism**

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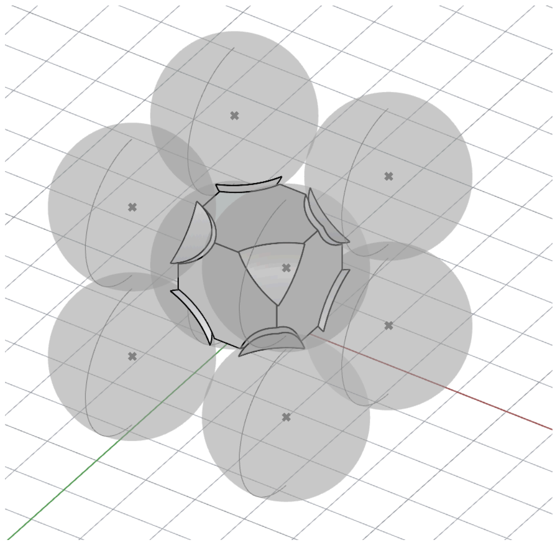
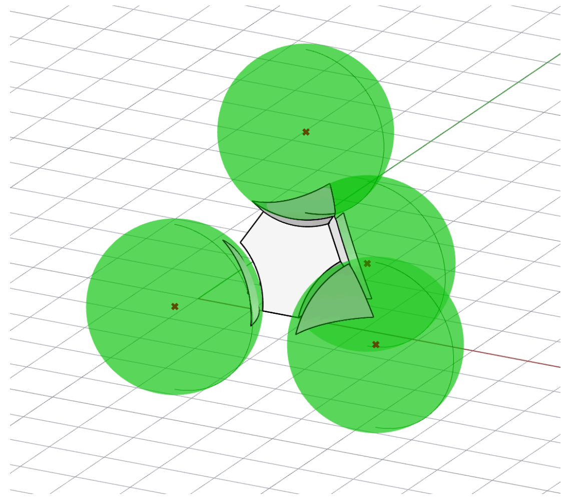
Variables to play with

- The **geometry** of the core

  How many inflatable modules can be combined - the core connector can be tetrahedron, cube, etc.



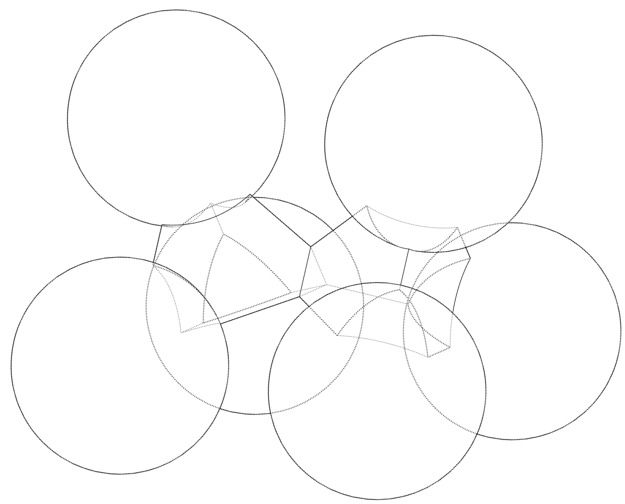


- ▼ **Shape-changing control mechanism**

  Learning based or timesteps/speed of the deflating and inflating, phase difference between the inflatable patches - now the demo using sine function to control the changing phases

- ▼ **Configurability (connections conditions)**

  reconfigurable structure - multiple units can be connected to each other opening up the space for different types of movement and behavior.



- **Weight**

  Balancing states - liquid pumped into different sections can change the center of the gravity.

- **Scale**

  How large the module can be?

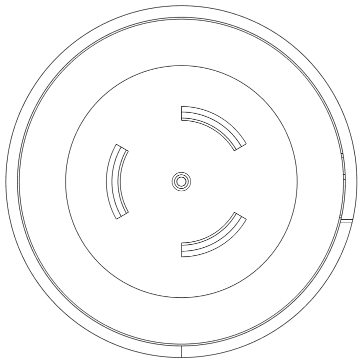
- **Material**

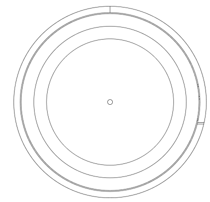
  The material such as silicone gel or polyethylene plastics has unique attributes that can contribute to the deforming process of the inflatable shape-changing parts.

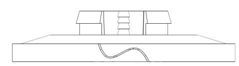
Physical Prototype

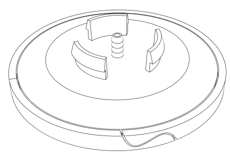
Structure Design

- **inflatable modular holder plate:**









▼ **Snap Fit Joint Design:**

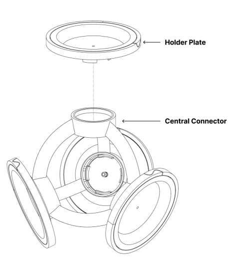
https://www.clarwe.com/blog/to-snap-fit-joints-types-design-and-manufacturing.html

https://hackaday.io/project/173560/logs sort=newest&page=2

https://www.madearia.com/blog/snap-fit-design/

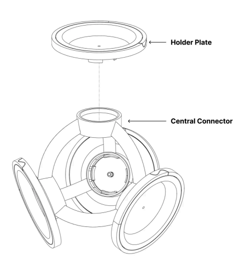
https://www.china-machining.com/blog/snap-fit-design-101/

▼ **Asembled structure example:**



Removable
modular design

▼ **Asembled structure example:**



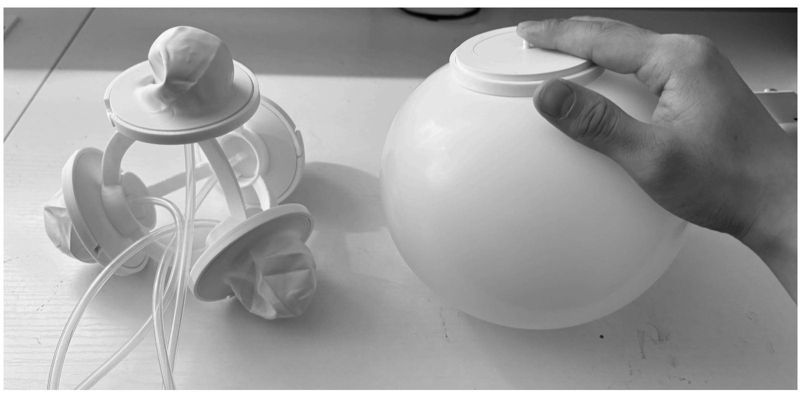
Removable
modular design

allowing developing components separately

allowing developing components separately

Preliminary findings:

1. There exists a maximum size the balloon can reach - the maximum inflated scale of the balloon compared to the whole skeleton



2. Active vacuum or passive leaking - deflating speed without additional active suction

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▼ Problem Findings:

1. **Sealing problem** - the air is easy to leak out on the edge and the tube connector
Solution or the 3d printed shell has some tiny unnoticeable cracks if it is too thin
2. **Uneven inflating speed** - fast in the beginning and gets slower proportionally as it inflates.
The time needed to reach the maximum size is around 40s.

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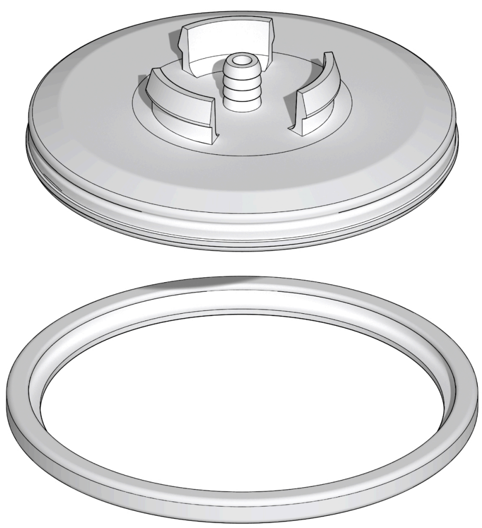
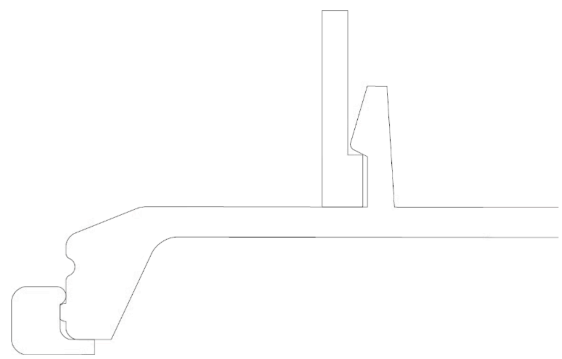
Iteration

Snap fit joint - from ring to cap

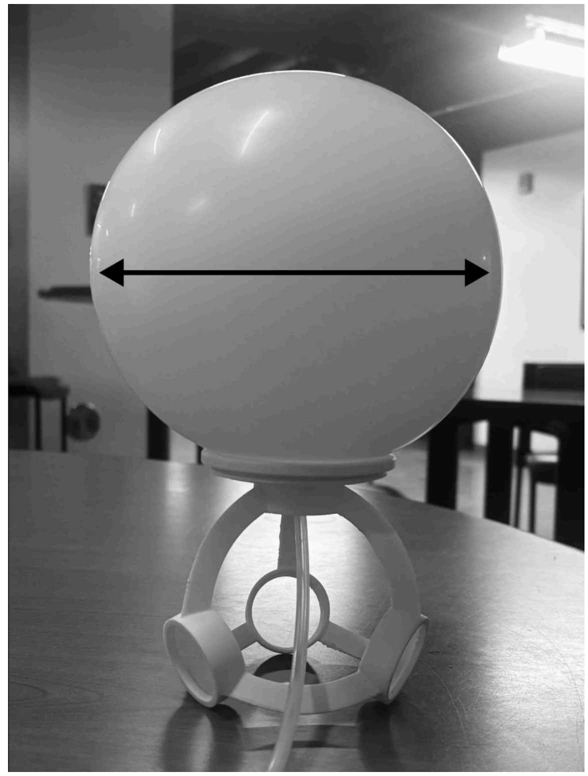


Better air tightness



Wriggle or Flip:

Vectors of the expanding - Horizontal expanding matters more than the vertical for flipping
(The size proportion between the central connector and the maximum inflated size.)



Internal balance:

The balloon has a tipping point. At the beginning, the rubber resists expansion and requires additional energy to stretch. Once one balloon passes this tipping point, it becomes easier for it to expand further.

As a result, the airflow follows the path of least resistance and preferentially flows into the balloon that has already expanded past the tipping point.

—Breathing Intelligence?

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—Separate the air tubing channels to
eliminate the influences?

Next Steps:

1. Go back to the simulation with higher fidelity to test out the different scale proportions of the inflatable modules and central structure. Verify the parameters that enable flipping gait.
2. Add constraint onto the inflatable module - morphological structure and shape-changing mechanism.
3. On-board vs. off-board

Questions:

Any other variables in this inflatable robot system you think I missed or value to experiment on?

Balloon, which is made out of elastic material (latex), has inflatable capacity, can be considered as a material and physic-based computational unit. By defining, controlling the connectability, constraints, the system ha

Hypothesis:
By controlling or setting up different connect